

QIHAO LIN

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Education

Northeastern University

Bachelor of Science in Computer Science — Concentration: Artificial Intelligence

Class of 2027

Boston, MA

Technical Skills

Programming Languages: Java, JavaScript, TypeScript, Python

Cloud/Database: AWS (S3, EC2, Bedrock, SageMaker, DynamoDB), Microsoft Azure, Firebase, MongoDB

Tools/Design: VS Code, GitHub, Jira, LaTeX, Raspberry Pi, Adobe Creative Suite

Experience

Objectways

Jan 2025 – Sep 2025

Software Developer Co-op

Phoenix, AZ

- Developed Tensoract, a data annotation platform processing 3M+ annotations, migrating the codebase from JavaScript to TypeScript + React, optimizing MongoDB schemas, and introducing AI-assisted labeling workflows.
- Implemented end-to-end testing with Playwright and integrated GitHub Actions CI/CD workflows to automate test execution and maintain code quality throughout development.
- Engineered a production management system integrating MongoDB APIs to automate attendance workflows for 2,000+ employees and developed JavaScript and React analytics dashboards for real-time workforce analytics.
- Deployed Sheetwise, a multi-LLM SaaS platform serving 200+ users, built on Azure, Firebase, Python, and JavaScript with Stripe integration and role-based access control for secure transactions.
- Developed AWS-based workflows for Amazon Q by researching and integrating services including Bedrock, Lambda, EC2, S3, DynamoDB, SageMaker, DocumentDB, API Gateway, Rekognition, Lex, and Connect; created technical documentation and training videos to enable team adoption.

University of Texas at Dallas

May 2021 – Aug 2021

Research Intern

Richardson, TX

- Quantified UAV vulnerabilities by simulating cyber-physical threat vectors on drone firmware with faculty.
- Programmed a Python-based Raspberry Pi controller that mitigated 100% of simulated DDoS attacks during field tests.
- Communicated research findings to 200+ peers and 7 faculty to advocate for enhanced autonomous security protocols.

Projects

Table Tennis Rally Analysis Tool | *Python, OpenCV, Computer Vision, Geometry*

May 2026

- Developed a table tennis rally tracker that follows the ball during play, detects where the ball lands on the table, and visualizes shot placement to help players analyze performance.
- Built a computer vision system that identifies the table's orientation from broadcast videos, even when table lines are not visible, by using geometric relationships and camera perspective analysis.
- Transformed ball bounce locations into a top-down table view, enabling analysis and visualization of shot placement and rally patterns.
- Engineered a unified full-frame coordinate system across corner detection, ball tracking, bounce detection, and map projection stages, eliminating offset-related errors and simplifying downstream processing.

SoloRed Card Game | *Java, OOP, JUnit*

April 2025

- Built a Java application to model game entities, player actions, and state management using object-oriented design.
- Designed classes for game entities, player actions, and rule evaluation to create a structured and maintainable codebase.
- Created comprehensive unit tests with JUnit to ensure reliability and correctness across diverse gameplay scenarios.

Leadership

Northeastern Table Tennis Team

Sep 2025 – Present

Team Captain

Boston, MA

- Secured 1st place in the 2026 National Collegiate Table Tennis Divisionals by directing team strategy and training schedules.
- Provided in-match coaching and tactical analysis, identifying opponent weaknesses and adjusting game plans to maximize team success.